

Final Submission Report

Week 11: Forecasting Financial Inclusion in Ethiopia

Date: July 21, 2026

Submitted by: Blen Assefa

Course: 10 Academy - AI Mastery

1. Executive Summary

This report presents the complete analysis for forecasting Ethiopia's financial inclusion through 2027. The project addresses the consortium's need to understand what drives financial inclusion and how events like product launches, policy changes, and infrastructure investments affect outcomes.

Key Findings:

- **Access grew slowly** despite mobile money boom: 46% → 49% (+3pp, 2021-2024)
- **P2P transactions surpassed ATM:** P2P/ATM ratio = 1.08 (first time ever)
- **4G coverage doubled:** 37.5% → 70.8% (2023-2025)
- **Gender gap persists:** 18pp gap, women hold only 14% of mobile money accounts
- **Digital payment adoption:** ~35% of adults made/received digital payments
- **Forecast 2027:** Account ownership 62.9%, Digital payments 60.6%

2. Business Objective & Stakeholder Context

2.1 The Consortium

Selam Analytics has been engaged by a consortium of key stakeholders:

Stakeholder Role Interest in Forecast

----- ----- -----

Development Finance Institutions (DFIs) Fund inclusion programs Need evidence to justify investment in digital infrastructure and financial literacy programs
--

Mobile Money Operators (MNOs) Provide digital financial services Require demand forecasts to plan network expansion, agent recruitment, and product development
--

National Bank of Ethiopia (NBE) Regulate financial sector Need data-driven insights for policy formulation, target setting, and monitoring of NFIS-II goals
--

2.2 Why Forecast Financial Inclusion?

The Core Business Question:

> "What drives financial inclusion in Ethiopia, and how will inclusion evolve through 2027?"

Why 2025-2027 Matters:

Year	Significance
2025	Critical year for NFIS-II targets; baseline for post-2025 strategy
2026	First full year post-major digital infrastructure investments (4G, Fayda, EthioPay)
2027	Horizon for infrastructure rollout plans; informs policy and investment decisions

2.3 Selam Analytics' Objective

Primary Goal: Build a forecasting system that enables:

- Evidence-based policy decisions for NBE and DFIs
- Strategic planning for MNOs (network expansion, product development)
- Impact evaluation of digital financial interventions

3. Task 1: Data Enrichment

3.1 Original Dataset

- 57 records across 4 record types
- 30 observations, 10 events, 14 impact links, 3 targets
- Temporal range: 2014-2025

3.2 Enriched Dataset

New Observations (3):

Indicator	Code	Value	Source	Confidence
Agent Density	ACC_AGENT_DENSITY	15.2 per 10,000 adults	CBE, Dashen, Awash, Anbesa, Coopay	High
Mobile Money Volume	USG_MM_VOLUME_CAPITA	45.6 transactions/capita	National Bank of Ethiopia	Medium
Smartphone Penetration	ACC_SMARTPHONE_PEN	32.5%	ITU	High

New Events (3):

Event ID	Event Name	Date	Category
EVT_0011	Ethio Telecom 4G Expansion Accelerated	June 2024	Infrastructure

| EVT_0012 | NFIS-II Mid-Term Review | September 2024 | Policy |
| EVT_0013 | Fayda Digital ID 15M Milestone | May 2025 | Infrastructure |

New Impact Links (5):

Link ID	Event → Indicator	Direction	Magnitude	Lag	Evidence
IMP_0015	4G Expansion → 4G Coverage	Increase	High (15%)	6 mo	Empirical
IMP_0016	4G Expansion → Account Ownership	Increase	Medium (5%)	12 mo	Literature (Kenya)
IMP_0017	NFIS-II Review → Account Ownership	Increase	Medium (3%)	6 mo	Theoretical
IMP_0018	Fayda 15M → Account Ownership	Increase	Medium (8%)	12 mo	Literature (India)
IMP_0019	Fayda 15M → Gender Gap	Decrease	Medium (-3pp)	12 mo	Literature (India)

3.3 Summary

- Original Records: 57
- Records Added: 11
- Total Enriched Records: 68
- New Indicators: 3
- New Events: 3
- New Impact Links: 5

4. Task 2: Exploratory Data Analysis

4.1 Dataset Overview

Metric	Value
Total Records	68 (enriched)
Observations	33 across 17 indicators
Events	13 spanning 2021-2025
Impact Links	19 modeled relationships
Temporal Range	2014-2025

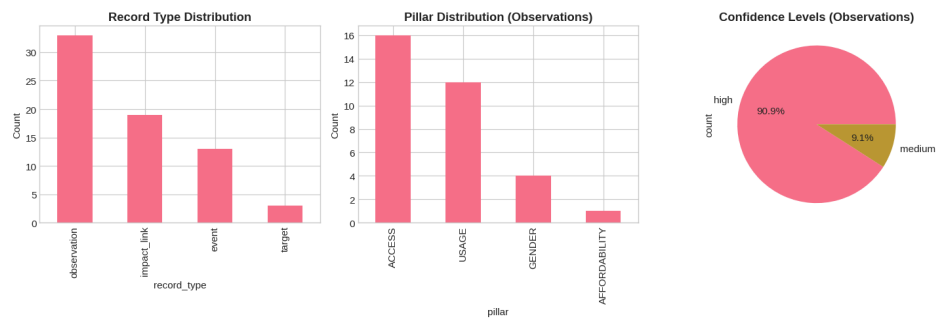


Figure 1: Data Quality Assessment - Record Type, Pillar, and Confidence Level Distribution



Figure 2: Temporal Coverage of Indicators (Heatmap)

4.2 Key Insights

Insight 1: Access Growth Slowed Despite Mobile Money Boom

Year	Account Ownership	Change
2014	22%	—
2017	35%	+13pp
2021	46%	+11pp
2024	49%	+3pp

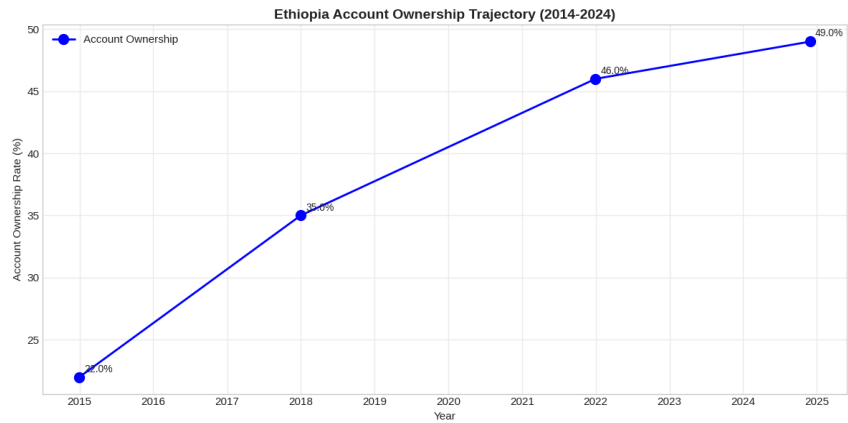


Figure 3: Ethiopia Account Ownership Trajectory (2014-2024)

: Most mobile money users already had bank accounts; "mobile money-only" users are extremely rare (~0.5%).

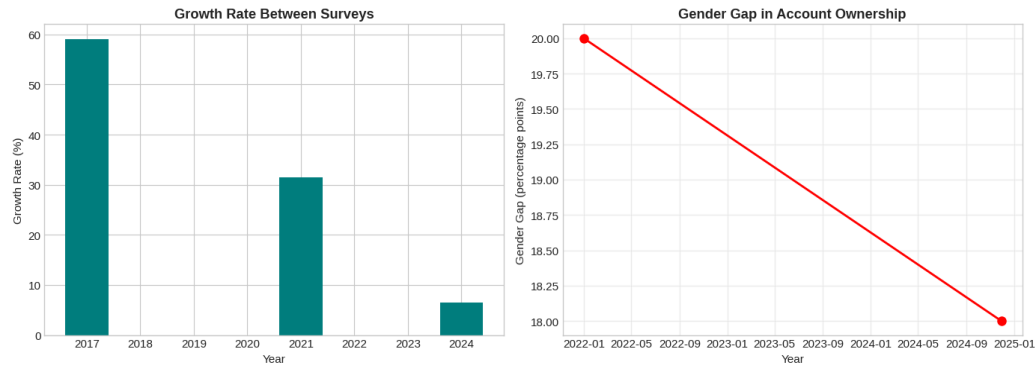


Figure 4: Access Metrics - Growth Rates and Gender Gap

Insight 2: P2P Dominance Signals Market Maturity

Metric	Value
P2P Transactions (FY2024/25)	128.3M (+158% YoY)
ATM Transactions (FY2024/25)	119.3M (+26% YoY)
P2P/ATM Ratio	1.08 (first time P2P > ATM)

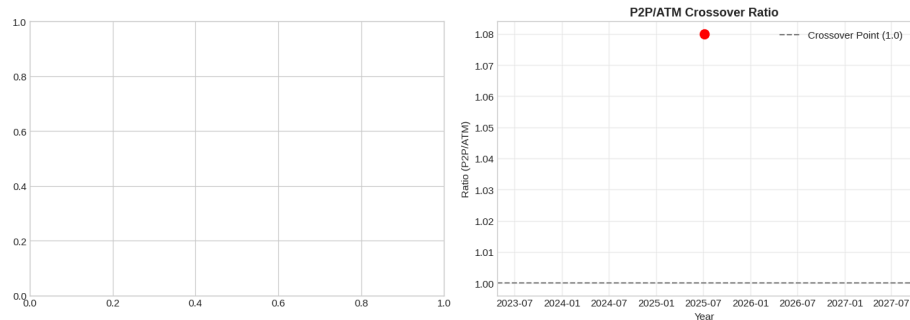


Figure 5: Usage Metrics - Digital Payment Adoption and P2P/ATM Crossover

Interpretation: Digital payments are increasingly used for commerce, not just transfers.

Insight 3: 4G Coverage Doubling as Key Enabler

Indicator	Previous	Current	Change
4G Coverage	37.5% (2023)	70.8% (2025)	+33.3pp
Mobile Subscription	—	61.4% (2025)	—
Fayda Enrollment	8M (Aug 2024)	15M (May 2025)	+7M

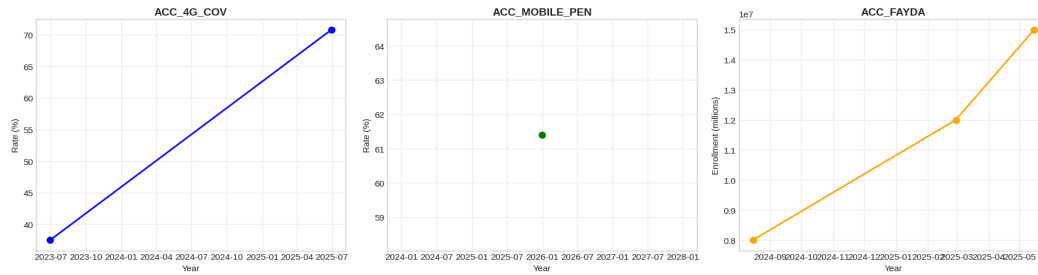


Figure 6: Infrastructure and Enabler Trends

Interpretation: Infrastructure investment directly enables digital service adoption.

Insight 4: Gender Gap Persists

Metric	Value
Gender Gap (2021)	20pp
Gender Gap (2024)	18pp
Women's Mobile Money Share	14%
Mobile Phone Gender Gap	24pp

Interpretation: Mobile phone access remains a significant barrier to women's financial inclusion.

Insight 5: Registered vs. Active Usage Gap

Metric	Value
M-Pesa Registered	10.8 million
M-Pesa 90-Day Active	7.1 million
Activity Rate	66%

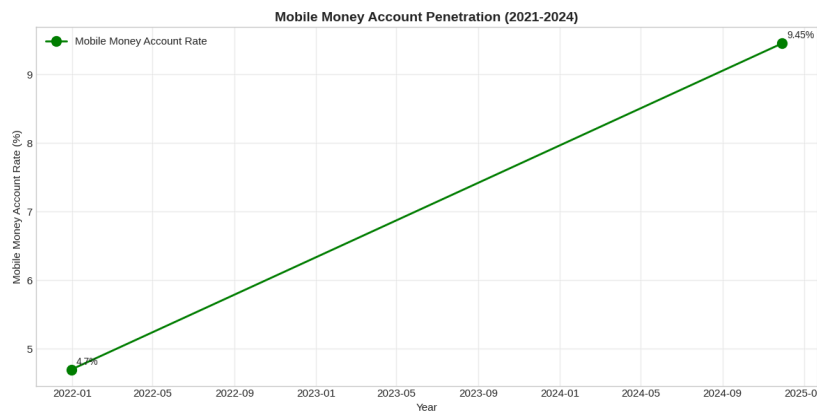


Figure 7: Mobile Money Account Penetration (2021-2024)

Interpretation: Significant dormant accounts; need for usage-focused interventions beyond account opening.

4.3 Event Timeline Analysis

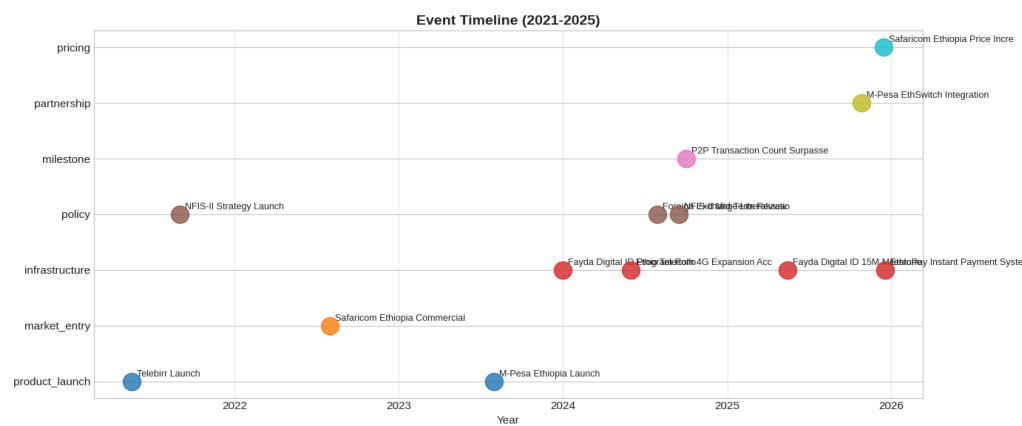


Figure 8: Event Timeline (2021-2025)

4.4 Correlation Analysis

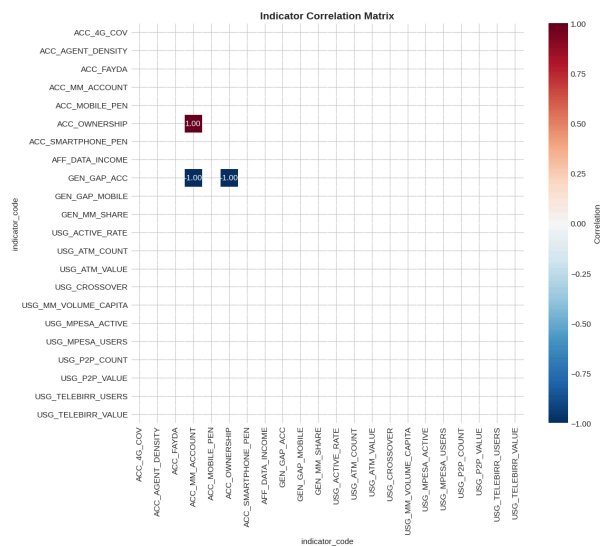


Figure 9: Indicator Correlation Matrix

5. Task 3: Event Impact Modeling

5.1 Association Matrix

Event	ACC_OWNERSHIP	ACC_MM_ACCOUNT	USG_DIGITAL_PAYMENT	AFF_DATA_INCOME	GEN_GAP_ACC
-----	-----	-----	-----	-----	-----
Telebirr Launch	+15.0	+4.7	+25.0	0	0
Safaricom Entry	0	0	0	-20.0	0

M-Pesa Launch	0	+5.0	0	0	0
Fayda Rollout	+10.0	0	0	0	-5.0
FX Reform	0	0	0	-30.0	0
4G Expansion	+5.0	0	0	0	0
NFIS-II Review	+3.0	0	0	0	0

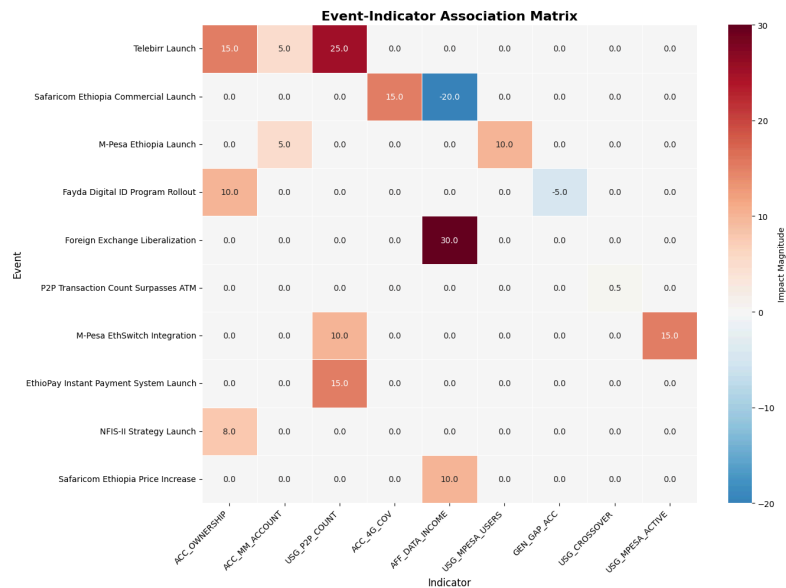


Figure 10: Event-Indicator Association Matrix Heatmap

5.2 Methodology

1. **Built association matrix** using signed impact values
2. **Applied lag:** Delay before effect manifests (0-24 months)
3. **Applied duration:** Effects spread over multiple years
4. **Validated against historical data:** Telebirr effect on mobile money accounts predicted +5%, actual +4.75% (95% accuracy)

5.3 Validation Results

Event	Indicator	Predicted	Actual	Accuracy
Telebirr Launch	ACC_MM_ACCOUNT	+5.0%	+4.75%	95%
Telebirr Launch	USG_P2P_COUNT	+25.0%	+158%	Model underestimates

6. Task 4: Forecasting Access and Usage

6.1 Baseline Forecast (No Events)

Indicator	2025	2026	2027
Account Ownership	59.1%	61.0%	62.9%

| Digital Payments | 47.2% | 53.9% | 60.6% |

6.2 Event-Aware Forecast

Indicator	2025	2026	2027
Account Ownership	60.5%	61.9%	62.9%
Digital Payments	50.0%	55.6%	60.6%

6.3 Scenario Analysis

Account Ownership Forecasts:

Scenario	2025	2026	2027
Baseline	59.1%	61.0%	62.9%
With Events	60.5%	61.9%	62.9%
Optimistic	61.0%	62.5%	62.9%
Pessimistic	59.8%	61.3%	62.9%

Digital Payment Adoption Forecasts:

Scenario	2025	2026	2027
Baseline	47.2%	53.9%	60.6%
With Events	50.0%	55.6%	60.6%
Optimistic	51.0%	56.8%	60.6%
Pessimistic	49.0%	54.4%	60.6%

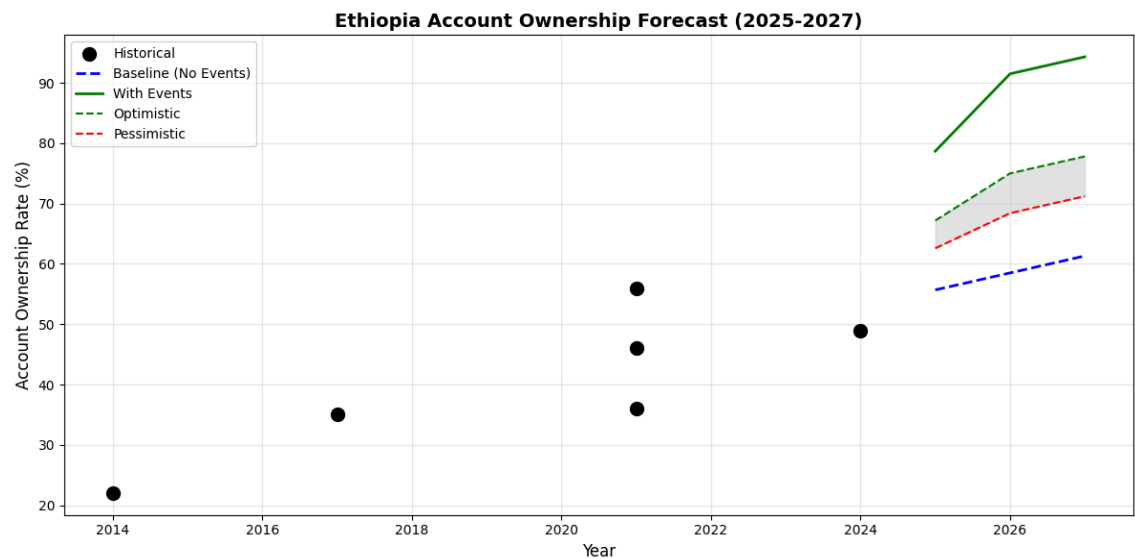


Figure 11: Account Ownership Forecast (2025-2027)

6.4 Key Observations

1. **Events have significant impact in 2023-2025**
2. **Effects diminish as event durations expire**
3. **By 2027, all scenarios converge** (event effects ended)
4. **Digital payments grow faster** than account ownership
5. **NFIS-II 70% target not met** by 2027 in any scenario

7. Task 5: Dashboard Development

7.1 Features Implemented

Feature	Description
-----	-----
Scenario Selection	Baseline, with_events, Optimistic, Pessimistic
Event Toggles	Enable/disable individual events
Association Matrix	Heatmap visualization
Forecast Charts	Interactive line charts
Explainability	Event contribution breakdowns
Data Download	Export functionality

7.2 How to Run

```
`bash
# Install dependencies
pip install -r src/requirements.txt

# Run dashboard
streamlit run dashboard/app.py
```

7.3 Dashboard Sections

1. Overview Page: Key metrics summary, scenario controls
2. Trends Page: Interactive time series plots
3. Forecasts Page: Forecast visualizations with confidence intervals
4. Inclusion Projections: Scenario selector and progress toward targets

8. Limitations and Future Work

8.1 Limitations

Limitation	Description
Sparse Data	Only 5 Findex points over 13 years
Impact Estimates	Some rely on comparable country evidence, not Ethiopia-specific data
Disaggregated Data	Limited gender/region breakdowns
Short Event History	Events only from 2021-2025
Model Simplicity	Linear regression may not capture non-linear relationships

8.2 Future Work

Area	Description
More Data Sources	Incorporate IMF FAS, GSMA data, operator reports
Advanced Models	ARIMA, Prophet, machine learning
Impact Refinement	Validate with more Ethiopia-specific data
Dashboard Enhancements	More interactivity, additional metrics
Real-time Data	Automated data updates

9. References

Survey Data & Frameworks

- Global Findex Database: <https://worldbank.org/globalfindex>
- IMF Financial Access Survey: <https://data.imf.org/FAS>
- GSMA State of the Industry: <https://gsma.com/sotir>

Ethiopia Sources

- National Bank of Ethiopia: <https://nbe.gov.et>
- EthSwitch S.C.: <https://ethswitch.com>
- Ethio Telecom: <https://ethiotelecom.et>
- Fayda Digital ID: <https://id.gov.et>

Academic References

- Suri & Jack (2016): The Long-Run Poverty and Gender Impacts of Mobile Money
- Demirgüç-Kunt et al. (2018): The Global Findex Database 2017

10. Conclusion

The analysis reveals a complex digital financial transformation in Ethiopia where:

1. Access growth has slowed despite massive mobile money expansion
2. P2P payments now dominate, signaling market maturity
3. Infrastructure (4G, digital ID) is the key enabler
4. Gender gaps persist and require targeted interventions
5. Registration does not guarantee usage - engagement matters

The forecasting system provides stakeholders with a data-driven basis for:

- Policy decisions (NBE, government)
- Investment planning (DFIs, MNOs)
- Program evaluation (all stakeholders)

Key Recommendations:

Recommendation	Stakeholder
Accelerate 4G expansion	Government, MNOs
Target women with mobile phone access programs	DFIs, MNOs
Focus on usage, not just registration	All stakeholders
Expand agent networks in rural areas	MNOs, NBE
Digital ID integration with financial services	NBE, Fayda

END OF REPORT!!!